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MAPPING THE MIND: A MODEL BASED ON THERAVADA BUDDHIST TEXTS AND PRACTICES

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ABSTRACT

We propose a functional model based on Theravada Buddhist texts and practices to show how the mind works in relation to both our senses and how we perceive the external world. Our model suggests that the mind acts as a common internal sense organ, receiving all sensory data from the five external senses. It shows how contact plays a central role in both the generation and continual reconstitution of feelings, perceptions and thoughts. This model suggests how previous memories can influence one's thoughts, by actively editing each cognitive pathway to create a distorted perception of what is experienced in the present moment. We also illustrate how individuals proliferate and cling to thoughts as a result of craving, and address how this can lead to suffering within the context of dependent origination. We believe this model could function as a basic conceptual map of the mind to facilitate a deeper phenomenological understanding of ongoing, complex mental interactions, within the context of Theravada Buddhist theory and practice.

Introduction

The core of Buddhist teaching, and indeed what is supposedly the Buddha's first discourse, can be found in the Four Noble Truths (Rāhula 1974). These are: (1) there is suffering¹, (2) there is a cause of suffering, which is craving; (3) there is cessation of suffering, with ending of craving; (4) there is a path to the cessation of suffering, which is the Noble Eightfold Path. Before proceeding to investigate the mechanisms involved in the generation of suffering, the Pali word *dukkha*, translated as suffering, needs some special consideration. According to John Peacock,

This word [*Dukkha*], together with many other words in Pali and Sanskrit, are ill-served by their English translations. The English rendering of both *dukkha/duḥkha* and many other primary Buddhist terms, fails to capture the many resonances found within the original languages. The principal meaning of the term '*dukkha/duḥkha*' is 'misery' or 'unpleasantness'. (Peacock 2008)

He goes on to suggest that 'if we require a one-word translation of the term *dukkha/duḥkha* then "unsatisfactoriness" comes the closest.' This linguistic clarity is of utmost importance, because as Peacock notes, investigation of *dukkha*, 'is the starting point of all Indian soteriology, not just of the Buddhist tradition' (Peacock 2008).

Although authors such as Carol Anderson have cautioned against an exclusive focus on the Four Noble Truths, stating that 'In studies of the four noble truths done outside of the countries that are primarily Buddhist in religious practice the reading audience is frequently unaware of the variety of teachings contained in Buddhist literature and/or of the various practices that are integral to Buddhist identities' (Anderson 2013), she nonetheless states that 'even in the face of the multitude of discourses, teachings, analyses and admonitions that are recorded in the Theravada canon, the four noble truths have emerged consistently as one of the most accessible teachings of the Buddha' (Anderson 2013). Thus, due to both its centrality and accessibility, our map of the mind will aim to provide greater clarity, through our proposed schematic model, on how to integrate the four noble truths into one's thought and practice in order to undo the substrate which gives rise to suffering.

The goal of Buddhist meditation practice is to cultivate a mind capable of overcoming suffering by abandoning what is identified as its substrate or root cause: craving. This is done through the systematic application of specific techniques, i.e. the Noble Eightfold Path² (i.e. the fourth Noble Truth), the eighth factor of this path being the cultivation of right concentration (Bodhi 2000, Samyutta Nikaya (SN) 56.11). However, it is important to note here that the ultimate goal of the Buddhist path is not merely the development of right concentration. There are two additional factors beyond these eight: right knowledge and right deliverance³ (also translated as right release) to be realised following the eightfold path (Nanamoli and Bodhi 2001, Majjhima Nikaya (MN) 117; Bodhi 2012, Anguttara Nikaya (AN) 10.103,105). These can be considered as 'fruits' of the path that accrue upon appropriate cultivation of the previous eight factors. However, it should be noted that although these practices go back to the time of Buddha, i.e. around 26 centuries ago, their modern use is derived from the nineteenth to twentieth century revival and promotion of meditation, primarily exported from Burma in the post-Independence period (Gombrich 2006).

According to the existing Theravada Buddhist literature, in his very first discourse, 'Setting in Motion of the Wheel of the Dharma' the Buddha explains the Four Noble Truths in detail. However, the word *dukkha* or suffering is not explicitly defined. In this discourse only examples of suffering are given. 'Birth is suffering, aging is suffering, illness is suffering, death is suffering; union with what is displeasing is suffering; separation from what is pleasing is suffering; not to get what one wants is suffering' (Bodhi 2000, SN 56.11). At the end of the examples however, it is said, 'in brief, the five aggregates⁴ subjected to clinging are suffering⁵ (Bodhi 2000, SN 56.11). This is one of the most important

statements in this *sutta*, but may not be easily or immediately understood by all readers. A potential reason for this may be that the mechanisms underpinning how individuals cling to the five aggregates are not explicitly defined in this *sutta*. However, if an individual is committed to undoing suffering, one needs to first understand this process of clinging clearly. The task or the duty of the first Noble Truth is to comprehend suffering. We therefore have to pay special attention to the mechanisms underlying habitual clinging to the five aggregates in order to uncover the true nature of the mental mechanisms underpinning the generation of suffering.

It is also stated in the *sutta* texts (Bodhi 2000, SN 55.5; Bodhi 2012, AN 10.61) that the Buddhist path begins through association with a superior person, such as an individual who is well-versed with the Dhamma (*sappurisasamsevo*). It should be noted that Dhamma cannot be understood as merely the verbal teachings of the historical Buddha, but as the natural course of phenomena and true nature of how they come to exist. Thus, even the historical Buddha did not 'discover' the Dhamma *de novo*, but 'uncovered' what is always already present and had been known by others before his time – this is covered in great detail in the *Nagara Sutta* (Bodhi 2000, SN 12.65). The other three factors important for progression along the path include: hearing true Dhamma (*saddhammassavanam*), careful attention (*yonisomanasikāra*) and practice in accordance with the Dhamma (*dhammānudhammappaṭipatti*) (Bodhi 2000, SN 55.5). The word *yonisomanasikāra* is another key concept in Buddhist practice, and has to be understood well. According to *Pali-English Dictionary* (Rhys Davids and Stede 1999), *yonī* is translated as the place of birth; *yoniso* means 'down to its origin or foundation.' *Manasikāra* simply means 'attention.' This in essence is the ability to be aware of the activity of the senses and the origin of mental phenomena, as it happens. The activity of the six senses⁶ and how we perceive and interpret sensory data will determine how we experience the external world. *Yonisomanasikāra* is translated as 'careful attention' (Bodhi 2000, SN 55.5; Bodhi 2012, AN 10.61), with the term 'careful' better understood as meaning 'pregnant' or imbued with care, rather than merely cautious or precise. Some have alternatively translated this word as 'appropriate attention' or as 'wise attention' (Olendzki 2013).

The Buddha is said to have once given some brief instructions on practice to a lay-person named Bāhiya (Nyanaponika and Hecker 2003), who is said to have become fully enlightened on contemplating the deeper meaning of that teaching with 'careful attention.' The very same instructions were given to an elderly monk called Venerable Maluṅkyaputta. The instructions given to both Bāhiya and Venerable Maluṅkyaputta were as follows:

Regarding things seen, heard, sensed, and cognized by you: in the seen there will be merely the seen; in the heard there will be merely the heard; in the sensed there will be merely the sensed; in the cognized there will be merely the cognized (Bodhi 2000, SN 35.95).⁷

It is said that both Bāhiya and Venerable Maluṅkyaputta contemplated this brief instruction, paying careful attention to the details of the statement and its deeper meaning. It is likely that they directly experienced the source or the origin of their cognitive phenomena playing out in their own minds, through the application of the mindfulness meditation methods conveyed in this brief instruction. Therefore, if one were to develop insight through meditation, it is important to direct careful attention to the origin where everything is born, in relation to the senses. Attentive observation of the origin or birth of phenomena is emphasised in *yoniso* (place of birth) of *yonisomanasikāra*, but this is lost in the English translation of this term as careful attention. *Vipassanā*,⁸ commonly known as insight meditation, is characterised by the application of *yonisomanasikāra* while contemplating the activity of the six senses, and the resultant five aggregates, as they arise and pass away moment-by-moment. As cognitive phenomena are rapidly and dynamically in flux, moment-by-moment, it is important to have a good knowledge of the flow of cognitive pathways of perceptions, feelings and thought formations when developing insight through meditation practice. It is difficult to understand how these cognitive processes work without proper conceptual knowledge, provided by the *sutta* texts that explain these cognitive pathways of perception. Unfortunately, no single *sutta* text explains all the necessary cognitive pathways in enough detail to understand the cause of suffering by clinging. However, many explanations are found scattered in many *sutta* texts throughout the Pali Canon.⁹ It is essential to make the necessary connections between these pathways, by putting together insights expressed in these various *sutta* texts, if one wishes to make clear sense of how the mind works and how each of us creates and perpetuates our own suffering.

Objectives and methods

The objective of this paper is to advance the understanding of how the mind functions using a schematic model based on Theravada Buddhist *sutta* texts. We therefore decided to conduct an in-depth study of Theravada Buddhist *sutta* texts from the Pali Canon to clearly understand these pathways of mental phenomena, and the process of clinging and suffering. We primarily used *sutta* texts translated by Bhikkhu Bodhi so that the translated Pali terms in referenced passages in this paper are consistent to the reader. This will also make the references easily accessible to the reader for further reading if necessary, by a series of similar publications called 'Teachings of the Buddha,' translated by Bhikkhu Bodhi, Bhikkhu Ñāṇamoli and Maurice Walshe. There are approximately 12,000 *sutta* texts in total available today (Kelly 2011). A large number of *sutta* texts address the cognitive pathways of perception, specifically discourses on mindfulness, the six senses, five aggregates and thought proliferation. We decided to restrict the focus of the paper to specific *sutta* texts that are directly useful to developing a model of the mind (Nanamoli and Bodhi 2001, MN 18, 28, 38, 137, 148, 149), specifically those

discourses on mindfulness, the six senses, five aggregates and thought proliferation. The main focus was to understand the mechanisms of clinging to help one comprehend fully how individuals create their own suffering by producing a schematic map of the mind and mental phenomena. However, we emphasise that this model is only meant to serve as a map, and accordingly the map should not be mistaken for the terrain. The richness and complexity of the terrain cannot be directly understood through our map, but like any good map, our model aims only to facilitate traversing a difficult and potentially dangerous terrain in order to arrive at a destination marked by freedom from the bonds of suffering.

We use our schematic model as a general template to explain some of the complex mental mechanisms of perception. We then address the cognitive pathways of perception that are activated when we see an object, applying our model to visual perception as a specific case, that can also be applied to any of the other senses. We then separated these complex cognitive mental phenomena into multiple steps to simplify an understanding of how we distort our perception, influenced by our previous memories. An additional goal was to address how we proliferate thoughts and cling to the five aggregates, constructing and clinging to a conceptual sense of self built around the five aggregates. This should enable the understanding and clear comprehension of how we, as individual 'selves,' create and perpetuate our own suffering. More importantly, the aim of this paper is to illustrate these very same mechanisms at work within the mind of the reader/practitioner, and to assist the cultivation of the requisite insight necessary to undo the perpetual cycles of craving that create and sustain the suffering of all sentient beings.

Our basic functional model

Our six senses (five external and one internal) create the perceptual world we live in. The six senses and their corresponding sensory inputs are called the *all* in Buddhist texts. 'And what, Bhikkhus, is the *all*?' 'The eye and forms (objects), the ear and sounds, the nose and odours (smell), the tongue and tastes, the body and tactile objects (touch), the mind and mental phenomena (mind objects). This is called the *all*.' (Bodhi 2000, SN 35.23).¹⁰ It is said in the texts 'by directly knowing and fully understanding 'the all,' by developing dispassion towards it and abandoning it, one is capable of destroying suffering' (Bodhi 2000, SN 35.27). As such, it is quickly revealed that 'the all' is not a metaphysical claim about 'what is,' but an epistemological proposition grounded in phenomenology: about 'what can be known' through 'what can be experienced.' Furthermore, familiarity with 'the all,' is not the end goal, but rather the destruction of suffering, or salvation, through cultivation of dispassion towards 'the all.' Thus, the fundamental premise being expressed is not of metaphysics, epistemology or phenomenology, but ultimately about the possibility of soteriology: a pragmatic doctrine culminating in salvation from suffering.

Before we can develop dispassion and abandon clinging to the activity of the six senses, we need first to investigate and understand how they function. For example, when the eye meets an object, the object is cognised by the brain. This is called 'eye-consciousness' in the Buddhist texts. Sense-consciousness 'comes to be in dependence on a dyad. In dependence on the eye and forms (objects) there arises eye-consciousness' (Bodhi 2000, SN 35.93, see also Nanamoli and Bodhi 2001, MN 28). Early Buddhism does not accept a 'pure-consciousness' but insists that all consciousness is 'consciousness of', i.e. depends on the dyad of receptor and object indicated here in relation to eye. This is reiterated in other *sutta* texts (Nanamoli and Bodhi 2001, MN 38) and the three elements of sense-consciousness, sense-object and sense-receptor are applied to all six senses. These specific sense-elements are: eye element, form element, eye-consciousness element; the ear element, sound element, ear-consciousness element; the nose element, odour element, nose-consciousness element; the tongue element, taste element, tongue-consciousness element; the body element, tactile-object element, body-consciousness element; the mind element, mental-phenomena element, mind-consciousness element (Bodhi 2000, SN 14.1).

Therefore, we can describe six types of sense-consciousness; eye-consciousness, ear-consciousness, nose-consciousness, tongue-consciousness, body-consciousness and mind-consciousness (Figure 2). Each sense-consciousness takes its type depending upon the requisite sensory dyad (e.g. eye and eye-object) that is present. Only one consciousness can be present at a given moment. For example, when eye-consciousness is present, there is no ear-consciousness (or any other types of sense consciousness) at the very same moment in time.

According to the Buddhist Psychological Model, attentional resources are limited: an individual can only be aware of one object at a time. The experience of a continuous stream of consciousness is produced by the rapid series of sense impressions and mental events arising and passing away, similar to how movement in a film is created by a rapid succession of individual still images. (Grabovac, Lau, and Willett, 2011)

Thus far, we have described the three elements pertaining to each sense, and how the coming together of each dyad gives rise to the respective sense-consciousness (Bodhi 2000, SN 35.93; Nanamoli and Bodhi 2001, MN 28). Once the necessary three elements are in place, for example the eye, the eye-object and eye-consciousness, 'the meeting, the encounter, the concurrence of these three things is called eye-contact'¹¹ (Bodhi 2000, SN 35.93). This is a necessary step for further visual events (e.g. perception, feeling and thought formations pertaining to the visual object) to occur (Figure 1). This process of contact applies to the other senses too, and is referred to as contact in Buddhist texts explicitly discussing this phenomenon. Those that do, do so in piecemeal fashion, but we have brought together the cognitive pathways mentioned in many different texts and provided schematic representation of them in the present article (Bodhi 2000, SN 12.2, 12.44, 35.28, 48.39; Nanamoli and Bodhi 2001 MN18, 38, 143, 148, 149).

Each of the six senses, with their respective sense-object, makes six types of sense-consciousness. Then, the meeting of the three, i.e. eye, object and eye-consciousness forms 'eye-contact'. The same process takes place with the other senses, too. Thus, there are six contacts generated from each of the six senses. However, in Figure 2, six types of contacts are for the sake of clarity depicted by one common contact. It should be noted that only one sense organ is active at any given moment. Thoughts generated by the cognitive processing of each of the five senses (external senses) as shown above are fed to the mind as mental objects. Thoughts can also be retrieved back from the mind as mind objects and form a new mind contact. In Figure 2, this is depicted by a short arrow-loop from the mind to mind objects.

Contact plays a central role in generating feelings, perceptions and thoughts expressed as our emotions. 'Contacted, one feels (*vedeti*), contacted one intends (*ceteti*), contacted one perceives (*sañjānāti*)'¹² (Bodhi 2000, SN 35.93). The word *ceteti* is translated here as intention, by Bhikkhu Bodhi. This word *ceteti* also means 'to think' or have thoughts or formulate volition. The Pali term *saṅkhārā* (one of the five aggregates) is also translated as 'volitional formations', mental formations or mental fabrications. In this model we have denoted *ceteti* and *saṅkhārā* which can both be understood as volitional in nature, as 'thoughts'. We also illustrate in the model that if there is no sensory contact, there is no

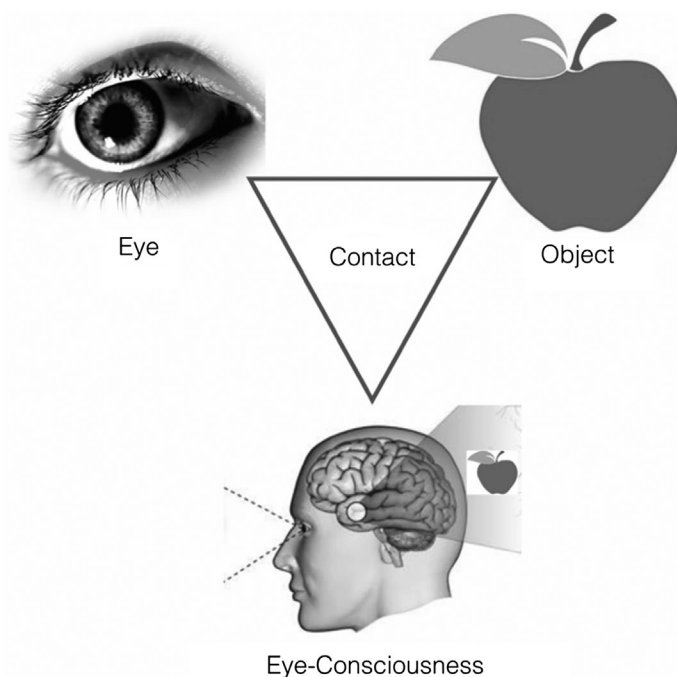


Figure 1. This figure shows the process of contact in the visual domain (i.e. eye-contact). Source: Adapted from Koch (2004).

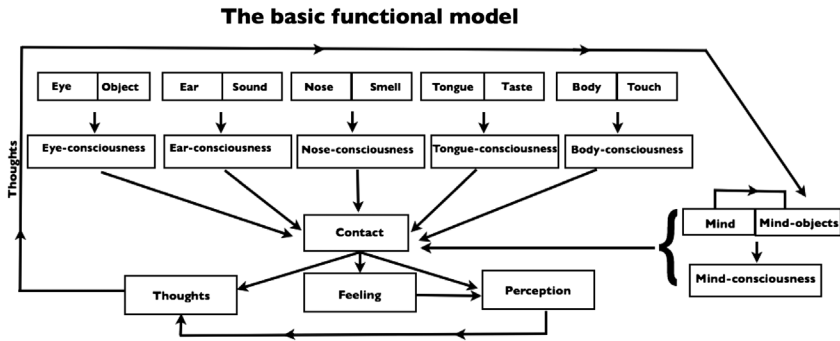


Figure 2. This figure shows the basic functional model. (In this model, as compared to Bhikkhu Bodhi's translations (Bodhi 2000, SN 12.2, 12.44, 35.28, 48.39; Nanamoli and Bodhi, 2001 MN38, 18, 143, 148, 149), we use the terms object for form, smell for odour, touch for tactile objects, and mind objects for mental phenomena.) Each of the six senses with their sense objects makes six types of consciousness.

Notes: For clarity, only one contact and subsequent cognitive event (i.e. thought, feeling or perception) is depicted. However, in actuality, there is a distinct contact for each of the six senses, but these contacts occur serially in time, even if they are commonly experienced as a unified multisensory field of conscious experience due to the rapid succession of fleeting mental events. Thoughts serve as mind-objects, and can arise either from the five sensory contacts or directly from the mind itself (e.g. as memories).

perception, feeling or thought process. This is consistent with all the Buddhist *sutta* texts we examined. In addition, *sutta* texts suggest that these cognitive flows are unidirectional. For example, as a result of contact, feeling occurs. Feeling does not directly give rise to contact, suggesting that this process does not operate in the reverse direction. This is also confirmed in the *sutta* texts. 'In dependence on the diversity of contacts arises the diversity of feelings. The diversity of contacts does not arise in dependence on the diversity of feelings' (Bodhi 2000, SN 14.5).

There are six types of sensory contacts that arise with each sensory encounter. Bhikkhu Bodhi translates it thus:

And how is it bhikkhus, that dependence on the diversity of elements there arises the diversity of contacts? In dependence on the eye element there arises eye-contact; in dependence on the ear there arises ear-contact; in dependence on the nose there arises nose-contact; in dependence on the tongue there arises tongue-contact; in dependence on the body there arises body-contact; in dependence on the mind there arises mind-contact. (Bodhi 2000, SN 14.2)

Only one type of sense contact can occur in a given moment. Therefore, at the precise moment when eye contact occurs, there is no ear contact (or any other form of sense contact). When we make eye contact, for example, related feelings, perceptions and thought constructs arise (Figure 2). This direction of the flow of mental information is suggested in the *sutta* texts. For example,

Dependent on the eye and the forms (object), eye-consciousness arises. The meeting of the three is contact. With the contact as condition there is feeling. What one feels, that one perceives. What one perceives, that one thinks about. What

one thinks about, that one mentally proliferates [about the object] (Nanamoli and Bodhi 2001, MN 18).¹³

The mechanism of mental proliferation will be discussed below (for a schematic illustration, see Figure 4). As with eye contact, the remaining senses also generate feelings, perceptions and thoughts. The Pali term for mental proliferation is *papañca*.¹⁴ Thanissaro Bhikkhu says, in translating *papañca*:

In Buddhist philosophical discourse it carries negative connotations, usually of falsification and distortion. The word itself is derived from a root that means diffuseness, spreading, proliferating. The Pali Commentaries define *papañca* as covering three types of thought: craving, conceit, and views. They also note that it functions to slow the mind down in its escape from *saṃsāra*. Because its categories begin with the objectifying thought, 'I am the thinker,' I have chosen to render the word as 'objectification,' although some of the following alternatives might be acceptable as well: self-reflexive thinking, reification, proliferation, complication, elaboration, distortion (Thanissaro 2014).

We will elaborate upon the mechanisms of mental proliferation using this model later in this paper. However, it is first important to understand the role that previous memory of mental events play in influencing mental proliferations in our basic model. We believe these previous memories are equivalent to what are known as *āsava* in Pali.

Buddhist *sutta* texts describe *āsava* as an important element responsible for defiling the mind. The Pali term *āsava* is translated to English as taint or mental corruption (Walshe 2012, Digha Nikaya (DN) 2). One *sutta* text gives a detailed description of how one should abandon taints (Nanamoli and Bodhi 2001, MN 2). While, it is not clear from this *sutta* how taints influence the cognitive flow of mental phenomena, the skills necessary¹⁵ to loosen the grasp of taints is described thus:

Bhikkhus, I say that the destruction of the taints is for one who knows and sees, not for one who does not know and see. Who knows and sees what? Careful attention and careless attention. When one attends unwisely, unarisen taints arise and arisen taints increase. When one attends wisely, unarisen taints do not arise and arisen taints are abandoned (Nanamoli and Bodhi 2001, MN 2).

Yonisomanasikāra, or careful/wise attention, as described earlier, returns as a means by which to attend to and thereby eliminate the impact of taints upon experience. By definition, taints are outflows and influxes to the mind and are considered as defilements of the mind (s.v. Rhys Davids and Stede 1999). The cognitive equivalent of the taints might be better understood as habits or memories. In our model, these previous memories are added to the influx of thoughts (see Figure 3). Thus, in our model, taints are understood as past memories of events that influence and modify our present feelings, perceptions and thoughts. There is an ongoing editing process with taints modifying conscious experience as we perceive and interact with the external world (ibid.). As the word implies, they literally taint or contaminate how we perceive the world. The word *āsava* translated as 'mental corruptions' would therefore capture this function more accurately than the translation of *āsava* as 'taints.'

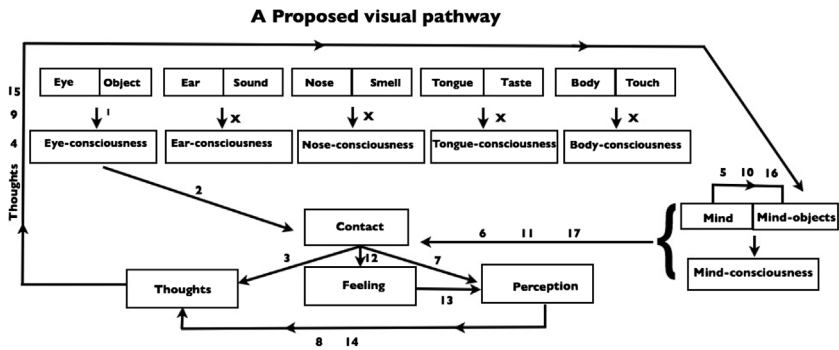


Figure 3. This figure depicts the cognitive flow of events that occur when an object is seen. Only a circuit of a single impulse is mapped here for the purpose of clarity.

Notes: The numbers from 1 to 17 are the possible initial sequence of events in the order that they are believed to occur. These are eye-consciousness (1), eye-contact (2), thought about the object (3–4), previous memory about the object (5), editing of the mind-contact (6), perception about the object (7), a thought about perception (8–9), previous memory related to the new thought (10), editing of another mind contact (11), feeling about the object (12), a further perception about the feeling (13), and another thought about it (14–15). A different previous memory (16) can edit the next mind-contact (17). This process can continue to proliferate after step 17 (not shown here). The schematic diagram illustrates how memories about the object (5, 10, 16) in the past can edit how we feel and perceive or think about the object, thus distorting how we see the object in any given moment.

The form (a body and its sense-receptors), feeling, perception, thought formations and consciousness are called the five aggregates in Buddhist texts (Bodhi 2000, SN 22.22, 22.48, 22.59; Nanamoli and Bodhi 2001, MN 109). Although we take the five aggregates to be constant entities, with the granularity of perception offered by careful inspection they could be better understood as transient activities of the mind. While one can more readily see how feelings, thoughts, perceptions and consciousness can be activities of the mind, it is not as easy to understand how physical form is also an activity of the mind. Form, more specifically its apprehension by human consciousness, although given as static, stable, and solid, is also in dynamic flux as the ongoing activity of perception. All five aggregates arise and pass away very rapidly, a result of their dynamic constitution. However due to the rapid velocity with which they arise and pass away, we perceive them as a constant flow or ‘a stream’ of activity of form, feeling, perception, thoughts and consciousness. Therefore, although we call these aggregates, they are merely activities with only varying degrees of persistent, rather than permanent, existence in one particular state or location.

The cognitive series of mental phenomena

When we see an object with the eye, there are already multiple mental loops or cognitive series that contribute to the richness of human perception. This is simplified and schematically illustrated here (Figure 3), using our basic functional

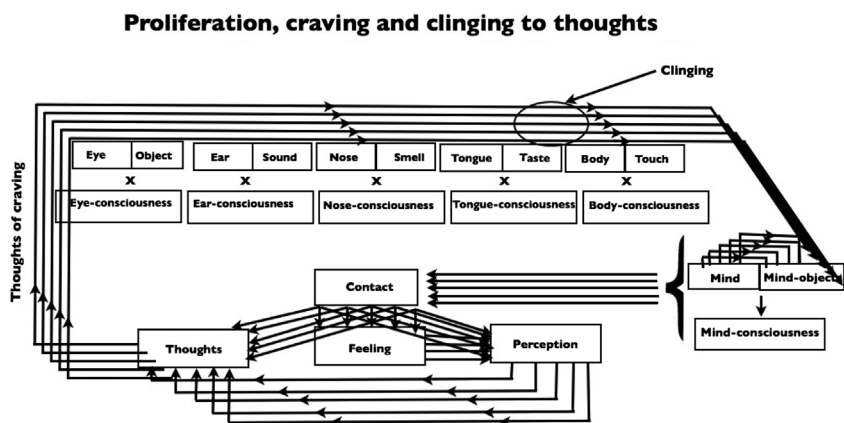


Figure 4. The figure above illustrates how we cling to our thoughts (e.g. a memory of an object). After we recognise the object by the eye through a series of cognitive processes (depicted in Figure 3), we can retrieve these as memories and form a series of multiple mind-contacts. The generation of multiple thoughts about the memory of the same object is called mental proliferation (cf. our discussion on *papañca*). These are multiple loops of thoughts of craving shown as multiple arrows, called sustenance, or clinging. In addition, previous memories can be added (at the level of the mind), to the mental flow of thoughts editing the present thought series (not shown in this diagram for the sake of simplicity, see Figure 3).

Notes: These multiple loops should be understood as strictly serial and dynamic cognitive processes. Here, they are illustrated as multiple static and parallel events for illustrative purposes only.

model as the template derived from Pali *sutta* texts (Figure 2). These cognitive series are important for us to get a comprehensive meaning about the object we are seeing, including the colour, shape, feelings and thoughts about the object. Since this occurs at a very rapid speed we are not able to discriminate these as single events in day-to-day activities. These events give us a distorted perception of an inextricably composite picture of a solid object. However, we have to realise that these multiple series of mental events are rapidly arising and passing away. These mental events are feeling, perception, contact, thoughts and attention (Nanamoli and Bodhi 2001, MN 9). These events are collectively termed *nāma* in Buddhist texts.¹⁶ Attention is not shown in the basic functional model above (Figure 2) or in the cognitive series (Figure 3). However, attention is better understood as a kind of thought (or an intention) redirecting and switching the activity of the cognitive series between the six senses very rapidly.

These cognitive processes of mental phenomena (*nāma*) very rapidly arise and pass away from moment to moment. *Nāma* in turn will trigger the activity of one member of the six sense dyads (e.g. eye or eye-object), or *rūpa*¹⁷, consequently giving rise to more *nāma*. Together they are called *nāma-rūpa*, translated as 'name-and-form' in Buddhist texts (Nanamoli and Bodhi 2001, MN 9, MN 38). This is how we perceive the external world through the six senses. With the development of insight, one observes these events taking place only

in the present moment. Even if we are thinking about an experience in the past or worrying about a future event, it all happens in the present moment. They are all activity of *nāma-rūpa* happening only in the present. The model (Figure 3) illustrates the expanded activity of *nāma-rūpa* and how *nāma* and *rūpa* interact. When we bring the well-trained mind to the present moment, we can observe all these events by applying careful attention with insight.

We will now examine the detailed activity of *nāma-rūpa* when we see an object, mapped as a cognitive series of a visual event (Figure 3, 1–17). First there is eye-consciousness (1). Note, the presence of eye-consciousness is a priori condition for the union giving rise to eye-contact (2): the union of eye, eye-object and eye-consciousness (Figure 1), not a serial step in each cognitive series (i.e. (1) has to have occurred in the past for (2) to be possible in any given cognitive series). The eye-contact will give rise to the first thought about the object (3–4). This will be merely a thought (4) that there is a visual object, but there is no information yet about a colour, shape or any feeling such as like or dislike about the object at this moment. This thought (4) will trigger a previous memory about the object (5). This will then make a mind-contact (6). This contact is edited by this previous memory and will give rise to a perception about the object, i.e. colour of the object (7). Thought (8–9) about this perception is then received by the mind as another mind-object. This will trigger another previous memory (10) related to the new thought that gives rise to another mind-contact (11). This contact too is edited by this previous memory. Then there is feeling (12) about the object i.e. likes or dislikes can further influence the next perception (13). For example, if we like the object we might appreciate the colour of the object more. Alternatively, if we dislike the object we might be less likely to appreciate the colour of the object. Then another thought (14–15) about this perception will occur. This time it may be the shape of the object. This will trigger another previous memory (16) about the shape of the object that can give rise to the next mind-contact (17). This contact too is edited by a previous memory. This process can continue to proliferate in a similar fashion as long as there is sufficient attention allocated to the object, allowing for further acquisition of sensory information about the object we have seen. Thoughts received as mind objects (4, 9, and 15) will be retained in memory, if they are significant for future use. This shows how previous memories about an object in the past can edit what we perceive, feel or think about the object in the present moment. This is how we distort our perception about an object, as is tainted by habit or memory (cf. our discussion on *āśava*), and provide a basis for the Buddhist position on the illusoriness nature of perception.

Alternatively, new information of the object you are seeing may trigger a previous memory and result in a cascade of higher-order cognitive loops or mental proliferations (depicted in Figure 4), dropping attention from the primary object and allocating it to secondary memories of that visual object, now in the form of mind-objects. This can result in a narrative pattern of thinking, using

only the internal sense base (i.e. mind and mind-objects) (Figure 4) which will be discussed in more detail further on in this paper. It is also possible to switch to another sensory cognitive pathway, e.g. the ear, very rapidly, depending on the next intention, e.g. if we want to listen to the sound coming from the object. It must be also noted that this entire cognitive process of the eye (Figure 3) will not occur with all the objects that trigger an eye-consciousness. Most will end at the first contact (1–2) but we do not become aware of these. Some may lead to the first mind object (1–4), but will stop there if the mind does not trigger a memory response to form a mind-contact (5). However, these numerous short cognitive pathways are also very important for our daily functioning, although we may not remember them well, as they do not pass through the mind multiple times (i.e. they do not persist as long, and increase the probability that it will be remembered in the present, or consciously apprehended). What we generally remember are the cognitive loops that are passed through the mind multiple times. These are especially the events that are significant to us and are thus what we tend to proliferate (Figure 4). They become habitual tendencies of our mind. *Sutta* texts support this idea. For example, ‘Bhikkhus, whatever a bhikkhu frequently thinks and ponders upon, that will become the inclination of his mind’ (Nanamoli and Bodhi 2001 MN 19). The idea that cognitive domains arise through a ‘process of circular causality (feedback systems)’ (Waldron 2002), supports our model of multiple feedback loops involved in the perception of an object. The nature of these feedback loops has also been succinctly described by Toneatto:

The cognitive phenomena of which we are aware are rarely based on an accurate or ‘true’ description of the environment as it presents to our six senses (the Buddha considered mind or consciousness to be a sense organ, with concepts, thoughts, etc., as its object). Rather, valid perceptions are inextricably bound up with our beliefs, feelings, assumptions, and attitudes about that perception. Thus, hearing a sound will not only lead to our auditory nerve being stimulated but also an array of conceptions about that sound (e.g., pleasant, loud, etc.). Similarly, the occurrence of a thought will usually elicit additional thoughts that elaborate, judge, or transform it and may be accompanied by images, memories, and somatic sensations, for example (Toneatto 2003).

How do we proliferate and cling to our thoughts?

The mind takes on an object (e.g. a memory) of a particular sense, followed immediately by another mind-object in the form desire, and then builds an identity of self around these and subsequent mental loops (a succession of which are referred to as mental proliferations) (Figure 4). Unskillful desires are called *taṇhā*, commonly translated as thirst or craving. This is identified as the main cause of suffering according to the Buddhist texts (Bodhi 2000, SN 56.11; Bodhi 2012 AN 6.63). ‘What is the source and origination of suffering? Craving is its source and origin’ (Bodhi 2012 AN 6.63).¹⁸ Our cravings are also thoughts (Bodhi 2012

AN 4.199) that arise as a result of feeling. Feeling will then give rise to craving and craving will in turn result in clinging according to dependent origination, a principle teaching on causality found in the Buddhist texts (Bodhi 2000, SN 12.2, 12.33, 12.61, 12.68; Nanamoli and Bodhi 2001, MN 28; Walshe 2012, DN 15).

Dependent origination is typically articulated as a twelve step causal process of mental phenomena¹⁹ and some important steps (i.e. steps one through eight) might be better understood using our proposed schematic model (specifically, Figure 4):

- *Step one of dependent origination:* ignorance serves as a supporting condition for volitional formation. Volitional formations are denoted as 'thoughts' in our model.
- *Step two of dependent origination:* volitional formations (or thoughts) serve as supporting conditions for the generation of consciousness. From our model, it is evident that the consciousness being generated is mind-consciousness; it is not any of the individual forms of sense-consciousness associated with the five external senses (i.e. eye-consciousness, ear-consciousness, nose-consciousness, tongue-consciousness or the body-consciousness).
- *Step three of dependent origination:* consciousness serves as a supporting condition in order to produce name-and-form. Feeling, perception, contact, thoughts and attention are grouped together as name-and-form in the *sutta* texts. Our schematic model shows how some of these components of name-and-form, namely feeling, perception, contact and thoughts interact. Furthermore, the model shows how contact can give rise to feelings, perceptions and thoughts. Attention, too, which is part of name-and-form is a thought. Attention will select the activity of one the six senses depending on the intention of a particular thought.
- *Step four of dependent origination:* name-and-form serves as a supporting condition for causing activity of one of the six senses. Thus, a new contact will be made.
- *Step five of dependent origination:* six sense bases (dyads of sense-receptors and sense-organs) serve as a supporting condition for contact.
- *Step six of dependent origination:* contact serves as a supporting condition for feeling.
- *Step seven of dependent origination:* feeling as a supporting condition will generate craving as the next step.
- *Step eight of dependent origination:* craving, too, is a type of thought. Multiple thoughts of craving lead to clinging, which is the next step of dependent origination.
- *Step nine of dependent origination:* clinging is perpetuated through multiple cycles of re-entry thoughts, with each thought feeding back and serving as a substrate, or mind-object, for the generation of another thought. This phenomenon is denoted as mental proliferation.

Note: Our model suggests that contact is involved twice within one cycle of the twelve-step dependent origination process. As dependent origination begins with ignorance (a thought), contact first occurs within the activity of the name-and-form (i.e. feeling, perception, contact, thoughts and attention) and thereafter the contact that is generated by the activity of the six senses (or sense bases). This implies that reentry (i.e. persistent cycling) of mental phenomena occurs at the level of contact in the cycle of dependent origination. Mental activity is always already present through the first contact, even before we make the second contact through one of the six sense bases. Importantly, it is said in the *sutta* texts that the steps leading to these dependently co-arising mental phenomena in dependent origination is the 'origin of this whole mass of suffering' (Bodhi 2000, SN 12.1, 12.2).²⁰ Therefore, activity of clinging in dependent origination is the first Noble Truth of the origin of suffering. Understanding dependent origination and seeing the interactions of its key components within oneself is the key to comprehending suffering.

According to dependent origination, feeling comes after contact. Feeling (tone) can be 'like' (positive affect), 'dislike' (negative affect) or 'neutral' (neutral affect) in response to a sensory object. Craving results in a proliferation of thoughts related to the memory of the sensory stimulus. This results in multiple re-entry thoughts as a consequence of the mind (i.e. the mind sense-receptor) making multiple new mind-contacts. This process of inflow of thoughts to the mind as mind-objects and outflow of memories as mind-consciousness continues as long as there is attention allocated to the memory of the object. This process is called mental proliferation (Figure 4). Although these cognitive pathways are shown as multiple parallel lines for the purpose of illustration (Figure 4), they only occur in a serial manner in time, as discussed above. It is said that:

Mental proliferation is simply a series of these mental events that has been triggered by an initial mental event or sense impression. Thus, in some cases, mental proliferation can feed on itself, with subsequent mental events having little to do with the sense impression that started the process. (Grabovac et al. 2011)

Another *sutta* text vividly illustrates these two contacts through the example of how we proliferate thoughts after a painful physical feeling.

Bhikkhus, when the uninstructed worldling is being contacted by a painful feeling, he sorrows, grieves, and laments; he weeps beating his breast and becomes distraught. He feels two feelings, a bodily one and a mental one. Suppose they were to strike a man with a dart, and then they would strike him immediately afterwards with a second dart, so that the man would feel a feeling caused by two darts. So too, when the uninstructed worldling is being contacted by a painful feeling, he feels two feelings, a bodily one and a mental one. (Bodhi 2000, SN 36.6)

The mental 'darts' can go on to proliferate, causing a third dart, a fourth, etc., resulting in self-perpetuating mental distress. Although this example is of a physical 'dart' shortly followed by a mental 'dart', this principle can also be applied to mental 'darts' that followed by subsequent other mental 'darts'. The mind can retrieve, for example, a traumatic memory of a previous event as the

first 'dart.' This can subsequently proliferate to produce multiple other mental darts (see short-arrow cycles from mind to mind-objects in Figure 4 onward). It is important to note, however, whether the eliciting or first 'dart' (i.e. contact) is of physical or mental origin, we are responsible for all the darts that we generate except for the very first 'dart' (i.e. contact). We generate our own mental suffering as a result of sustaining its proliferation. The unique premise of Buddhist soteriology is that each of us also possesses within ourselves the possibility of cutting off the sustenance for this proliferation, abolishing the fuel that feeds the fire of our own suffering.

When it is said that the 'eye is burning' (a similar metaphor may apply to each of the other senses) in the *sutta* texts (Bodhi 2000, SN 35.28), this means that after an object is perceived by the eye, the fire of craving ignites to set off the process of mental proliferation. Multiple thoughts of craving will result in what is called clinging (Bodhi 2000, SN12.52, SN 22.122). The Pali term for this is *upādāna*.²¹ *Upādāna* can also be translated as sustenance, to nourish or to feed. To sustain its activity, the mind needs nutriment. There are physical and mental nutriment described in the Pali Canon. 'There are four kinds of nutriment for the maintenance of beings that have already come to be and for the support of those seeking new existence' (Nanamoli and Bodhi 2001, MN 9; Bodhi 2000, SN 12.11). The four nutriment are described as physical food, contact, mental volition (thoughts) and consciousness.²² The first is the physical nutriment and the other three are mental nutriment. Contact plays a central role in providing the first mental nutriment and generating the other two nutriment, namely thoughts and consciousness. As shown in the model (Figure 2), thoughts are received by the mind as mind-objects, which together become mind-consciousness. This becomes the consciousness nutriment for further mind activity (Figures 3 and 4), the fuel required for the fire of mental activity to continue and proliferate. In one *sutta* text, a metaphor is used to explain how the mind consciousness grows – it is compared to a growth of a plant from five different seeds (Bodhi 2000, SN 22.54). The five types of seeds are compared to the five types of consciousness that arise from the five senses. Mind consciousness is compared to a 'plant' here and it grows supported by the other four aggregates (form, feeling, perception, thoughts) as the 'earth.' The craving here is compared to the 'water' essential for the growth of the 'plant.' In another *sutta* text (Walshe 2012, DN 33), the four aggregates; form, feeling, perception and thoughts, are said to be the four stations for consciousness²³; where it grows, increases and flourishes.

Thoughts taken as nutriment can also further propagate the five aggregates. The definition of thoughts (volitional formations²⁴) in a *sutta* confirms this.

And why, bhikkhus, do you call them volitional formations? 'They construct the conditioned,' bhikkhus, therefore they are called volitional formations. And what is the conditioned that they construct? They construct conditioned form as form; they construct conditioned feeling as feeling; they construct conditioned

perception as perception; they construct conditioned volitional formations as volitional formations; they construct conditioned consciousness as consciousness. 'They construct the conditioned,' bhikkhus, therefore they are called volitional formations. (Bodhi 2000, SN 22.79)

There is perpetual generation of the five aggregates by the thoughts, and this is shown in our models (Figures 3 and 4). It is interesting to note the mention of 'conditioned,' forms, feelings, etc., making new forms, feelings. The word 'conditioned' may be understood to mean 'edited by previous memories' as described before (Figure 3). In Buddhist terminology they are likely to be equivalent to *āsava* as we have discussed above.

We now know how we cling to the five aggregates with thought. However, it is not immediately clear why this clinging causes us to suffer. What we are doing essentially is clinging or 'feeding' the mind with the thoughts of sensual desires. The pleasure of feeding is always around the five aggregates (Bodhi 2000, SN 22.48). We feed on pleasant forms (objects), pleasant feelings, pleasant perceptions, pleasant thoughts and pleasant ideas for cognition. In other words, we keep generating more and more aggregates around the particular sensual object we desire. However, one does not realise that all these aggregates are transient in nature, and are thus impermanent (Bodhi 2000, SN 22.79). They arise and pass away swiftly and this impermanence of a desired pleasant experience causes us to suffer. This is because we expect them to be permanent and we desire to keep enjoying the gratification of these pleasurable aggregates. So we cling to them, leading us to suffer when we realise they are no longer there (or will no longer be there in the future) for us to enjoy. In the *sutta* texts, this is summarised by the statement, 'what is impermanent is subject to suffering' (Bodhi 2000, SN 22.15, SN 22.59).²⁵ Impermanence in Buddhism is understood as one of three marks of existence (the other two being suffering and non-self).

It is also said in the *sutta* texts that 'when one dwells contemplating gratification in things (five aggregates) that can be clung to, craving increases. With craving as a condition, clinging comes to be' (Bodhi 2000, SN12.52). In this *sutta* text, gratification is compared to a man adding more dry wood and other burning material from time to time to a great bonfire that is already burning. Essentially, we end up fuelling the aggregates further (making more aggregates) with fires of craving (Bodhi 2000, SN 35.28). It is also said 'when one dwells contemplating danger in things (five aggregates) that can be clung to, craving ceases. With the cessation of craving comes cessation of clinging' (Bodhi 2000, SN12.52). This is compared to a man who would not put dry wood and other burning material to a great bonfire that is already burning. It is said, 'When the former supply of fuel is exhausted, that great bonfire, not being fed with any more fuel, lacking sustenance, would be extinguished' (Bodhi 2000, SN12.52). According to another *sutta* text, one who has trained in the path, 'extinguishes and does not kindle' the five aggregates (Bodhi 2000, SN 22.79).

Clinging to a distorted perception of self

According to Buddhist *sutta* texts, we develop a distorted perception of a persisting and permanent self, leading us to become attached to our bodies, its feelings, perceptions, thoughts and consciousness. This clinging causes a distorted identification of a permanent self. There is clinging to the body, feelings, perceptions and consciousness as me, mine or as my-self (Bodhi 2000, SN 22.59; Nanamoli and Bodhi 2001, MN 11). McWilliams nicely summarises this process of thoughts leading to the construction a self and the construction of a self leading to further reification of thoughts:

Perception leads to thoughts, which proliferate, distorting experience. This proliferating cognitive tendency leads perceivers to construct sensory experiences and socially constructed ideas and beliefs about reality. Sensory contact begins a process that fabricates biased inferences, leading to concepts and views of objects. The construction of self (I, myself, mine) arises from this process of conceptual proliferation. Viewing thoughts as 'mine,' we treat them as real. (McWilliams 2011)

All these constructs are transient in nature, arising and passing away very rapidly, as we have discussed previously. However, because this turnover is extremely rapid, we are liable to see a dynamic process as a static 'collection' or aggregate: constant, pleasurable, and self-like in nature. This is because of our distorted perception of reality, due to a lack of the careful attention required to identify and observe mental processes. This deficient state, lacking in the careful attention necessary to see things as they are, is known as 'ignorance'.

Ignorance in Buddhist texts is not the lack of intellectual knowledge, but a matter of not realising the true nature of mental phenomena and how we create our own suffering. Craving mainly causes a clinging to self, as 'this is mine'. So we cling to the aggregates saying this is mine (Bodhi 2000, SN 22.59). A dialogue between the Buddha and his disciples emphasises this point:

'What do you think, bhikkhus, is form permanent or impermanent? What do you think, bhikkhus, is feeling permanent or impermanent? What do you think, bhikkhus, is perception permanent or impermanent? What do you think, bhikkhus, is volitional formations permanent or impermanent? What do you think, bhikkhus, is consciousness permanent or impermanent?' – 'Impermanent, venerable sir.' – 'Is what is impermanent suffering or happiness?' – 'Suffering, venerable sir.' – 'Is what is impermanent, suffering, and subject to change fit to be regarded thus: 'This is mine, this I am, this is my self'?' – 'No, venerable sir.' (Bodhi 2000, SN 22.79)

Another *sutta* text clarifies this point further. 'What is impermanent is suffering. What is suffering is nonself. What is nonself should be seen as it really is²⁶ with correct wisdom thus: 'This is not mine, this I am not, this is not my self.' (Bodhi 2000, SN 22.15).

Ignorance causes us to believe the five aggregates constitute a permanent self, taken as 'I am.' Clinging to an illusory self, as 'this is myself', is mainly due to a wrong view about a permanent self. Thought, speech and action in the service of desire and craving satisfy a fictive 'me' within. This is called the self-identity

view or the personality view (*sakkāyadiṭṭhi*) in the Buddhist literature. With each experience one makes multiple copies of self-identities (i.e. attributing selfhood and ownership to the body, objects and the environment) mistaking the five aggregates for the self. This is compared to a dog tied to a post by a leash in the *sutta* texts (Bodhi 2000, SN 22.99). The dog can only circle around the post; the dog is never let loose from it. The leash is taken to represent views and the post to represent personal identity (mistakenly built around the five aggregates). Tied to the post of personal identity by a leash of views, the uninstructed person has no recourse but to run ceaselessly around in circles until the views that bind them are somehow cut. Instruction into the illusoriness of personal identity frees the individual who was once uninstructed and lamely bound. Clinging to the five aggregates, deceived by wrong views, only serves to reinforce one's personal identity (Nanamoli and Bodhi 2001, MN 109). One way to transition from uninstructed to instructed, is through the instruction (or insight) provided by *vipassanā* meditation practice that reduces clinging. If we let go of all clinging, we are free from suffering and stress. This is called release. As Mikulas describes:

Vipassanā meditation is often called 'insight meditation' because it is held that cultivation of mindfulness leads to a form of insight called *prajna*, an immediately experienced intuitive wisdom. *Prajna* involves mindful and penetrating seeing into the fundamental nature of things in a way that transforms one's being, the ultimate purpose of meditation in Buddhism. For example, insightful seeing of impermanence [or suffering, or non-self] leads to a reduction of clinging. (Mikulas 2011)

Discussion and summary

We proposed a map to explain how the mind works using a schematic model based on Theravada Buddhist texts and practices. We illustrated how craving, clinging and mental proliferation works, and how we cling to an illusion of self formed around the five aggregates. Internal investigation of thought processes is critical to contemplative traditions like Buddhism. What is cultivated through such internal investigation is called *yonisomanasikāra* in Pali *sutta* texts, and is commonly translated as careful attention. In this paper, we have laid out a model based on Theravada Buddhist texts that one can use as a map to examine their own mind with careful attention. After all, the entire purpose of meditation practice is to see the inner workings of the mind and understand how we create our own suffering. Thus, we hope this model will facilitate such an effort by visually representing the elements and causal relations or processes involved.

Many readers of *sutta* texts understand dependent origination as a 12 step formula, at a largely theoretical level. However, it is important to realise these mental phenomena and their causal links at an experiential level. Here, we have demonstrated some key steps of dependent origination and how they can be schematically illustrated (Figure 4) in a manner amenable to individual experiential examination. With contemplation and insight, we can realise these are

merely a chain of mental phenomena that arise and pass away very rapidly and it is not necessary or beneficial to cling to them as self. It is said in a *sutta* text that one who has seen well with right discernment these dependently arisen mental phenomena will not run after the past or the future, nor will he be perplexed in the present moment about existence of a self (Bodhi 2000, SN 12.20).

It should be stressed that contact plays a crucial role in the formation of the cognitive flow of mental phenomena, serving as a key nutriment for the mind. Contact may be considered as 'the gateway' we can be aware of, with right mindfulness. Mindfulness, however, is not the same as careful attention. The right mindfulness in the Noble Eightfold Path is often misunderstood in the West. However, a detailed analysis of this misconstrual is beyond the scope of the present paper. Nonetheless, we note that right mindfulness should be defined as remembering to be skilful in the present moment. Mindfulness and clear comprehension are often grouped together in the Pali Canon as mindfulness-clear comprehension.²⁷ A *sutta* text explains the meaning of this term.

What is the development of concentration that leads to mindfulness and clear comprehension? Here a bhikkhu knows feelings as they arise, as they remain present, as they disappear; he knows perceptions as they arise, as they remain present, as they disappear; he knows thoughts as they arise, as they remain present, as they disappear. (Bodhi 2012 AN 4.41)

This suggests clear comprehension also involves a degree of discernment or wisdom. The factor responsible for mindfulness and clear comprehension is considered to be careful attention: 'Mindfulness and clear comprehension, too, I say, have a nutriment; they are not without nutriment. And what is the nutriment for mindfulness and clear comprehension? It should be said: careful attention' (Bodhi 2012 AN 10.61). According to Bhikkhu Bodhi:

The two terms, *sato* and *sampajāno*, often occur in proximity, implying a close affinity between their respective nouns, *sati* or mindfulness and *sampajāñña* or clear comprehension. To distinguish the two, I would describe mindfulness as lucid awareness of the phenomenal field. This element of lucid awareness prevails in the initial stages of the practice. But with the strengthening of mindfulness, clear comprehension supervenes and adds the cognitive element. In the practice of insight meditation, the meditator clearly comprehends the nature and qualities of arisen phenomena and relates them to the framework defined by the parameters of the Dhamma, the teaching as an organic whole. The expression 'clearly comprehending' thus suggests that the meditator not only observes phenomena but interprets the presentational field in a way that sets arisen phenomena in a meaningful context. As the practice advances, clear comprehension takes on an increasingly more important role, eventually evolving into direct insight (*vipassanā*) and wisdom (*paññā*). (Bodhi 2011: 22)

Clarification of the intimate relationship between mindfulness and careful attention has been referenced in the literature (Schwartz 1999). It is important to note that, according to Bhikkhu Bodhi:

Yoniso manasikāra is not synonymous with mindfulness, though the two work closely together. *Yoniso manasikāra* is close and careful consideration of things

taken up for scrutiny. Mindfulness is what makes it available for consideration, but *yoniso manasikāra* goes beyond 'Bare Attention' and actively examines things in the way intended to bring their real characteristics to light. The product of *yoniso manasikāra* is wisdom (*paññā*). Thus we might consider *yoniso manasikāra* to be the bridge between *sati* (mindfulness) and *paññā* (Schwartz 1999).

With the use of *yonisomanasikāra*, it is possible to focus our awareness upon rapidly changing contacts as they are generated from our six senses during meditation practice. We could use contact as our central awareness and see them arise and pass away, revealing their impermanence without generating any feeling of craving or aversion towards it. 'Bhikkhus, eye contact is impermanent, changing, becoming otherwise'²⁸ (Bodhi 2000, SN 25.4). 'When touched by pleasant contact do not be enthralled, do not tremble when touched with pain. Look evenly on both pleasant and painful, not drawn or repelled by anything' (Bodhi 2000, SN12.52). 'When the nutriment contact is fully understood, the three kinds of feeling are fully understood. When the three kinds of feeling are fully understood, I say, there is nothing further that a noble disciple needs to do' (Bodhi 2000, SN 12.63). Understanding the casual relation between contact and the resultant feelings (the most powerful subsequent causal factor leading to clinging) will reduce or inhibit our thought proliferation and clinging. This is confirmed in a *sutta* text. 'With remainderless fading away and cessation of six bases for contact there is cessation of proliferation, the subsiding of proliferation' (Bodhi 2012 AN 4.173). It is also said that 'when common people of proliferated perception, perceive and proliferate they become engaged' (Bodhi 2000, SN 35.94). The 'goal (of the path) is for one who does not delight in and enjoy proliferation, not for the one who delights in and enjoys proliferation'²⁹ (Nanamoli and Bodhi 2001, MN 11). We can gradually reduce our attention to rapidly changing contacts as it is generated from our six senses during meditation practice. It is said in the *sutta* texts, 'Bhikkhus, these six bases for contact – if untamed, unguarded, unprotected, unrestrained – are bringers of suffering'³⁰ (Bodhi 2000, SN 35.94). It is likely that the brief instructions given by the Buddha, 'regarding things seen, heard, sensed, and cognized by you: in the seen there will be merely the seen; in the heard there will be merely the heard; in the sensed there will be merely the sensed; in the cognized there will be merely the cognized' (Bodhi 2000, SN 35.95), is an attempt to encourage the individual to stop the cognitive series at contact. In other words, if you can train yourself through meditation to stop at seen, heard, sensed (i.e. smell, taste, and touch) and cognized (memories by the mind), you can lessen the burden of suffering. However, this will take a great deal of meditation practice for someone to be able to apply this in their everyday life. Until this is achieved, at least being aware of the mental proliferative activity of the mind, even at a theoretical level, will help one to disengage more quickly from the deleterious narrative patterns of thinking, and this will incrementally free the mind from suffering.

Conclusion

With our schematic map of the mind, we have attempted to simplify a very complex and dynamic mental process, described in a large number of Theravada *sutta* texts, in order to make this process more accessible to the wider Buddhist community and others who might be interested in what these teachings have to offer. More importantly, we emphasise that interested individuals can use this mind map as a tool in their personal meditation practice, as discussed above. Only through personal experience, with *sutta* texts and models such as those offered in this paper as an assistive tool, can we test how these mental processes play out in our own minds and eventually free ourselves from suffering. This happens when we know and see for ourselves that suffering is generated in our own minds, due to our own ignorance, due to an inability to apply careful attention. Contact should be regarded as a key component in the mental process that we can be aware of, contemplate, and develop insight regarding their arising and passing away in real time. Then perhaps one day, when we see, hear, smell, taste, touch or think, we can pause, contemplating these events as merely 'seen, heard, sensed and cognized', without getting caught up in habitual and proliferative patterns of thinking. We are essentially pausing, or ideally stopping, the cognitive series at the level of contact. This can immediately lessen the burden of suffering when applied practically. In other words, one is no longer adding any 'fuel to the fire' of suffering. Instead, one is extinguishing the fire where it started, at its source or origin. When we fully comprehend the underpinnings of the mental activity of our own mind, we can comprehend the genesis of suffering. For this, we need to understand clearly the mechanisms of how these complex cognitive phenomena work, and we hope this conceptual model will be a useful framework to achieve this goal.

The goal (of the path) is for one who does not delight in and enjoy proliferation, not for one who delights in and enjoys proliferation. – Buddha (MN 11)

Notes

1. Some authors have alternatively translated *dukkha* as 'stress' or 'unsatisfactoriness', to mitigate the profoundly negative tone implied by the word 'suffering' and to maintain even the experience of the most subtle dis-ease in their definition.
2. The Eightfold Path is comprised of eight factors: right view, right intention, right speech, right action, right livelihood, right effort, right mindfulness and right concentration. The word 'right' is found as a prefix for each of these path factors. This word signifies a specific technique, producing a desired result, rather than 'right or wrong' in a rule-based moralistic sense. Thus, insofar as it relates to ethical action, it is 'right' in the contextually governed, rather than rule based (i.e. deontological), sense.
3. Pali: *sammāñāṇaṃ* and *sammāvimutti*.
4. Form, feeling, perception, volitional formations and consciousness (Pali: *rūpaṃ, vedanā, saññā, saṅkhārā*, and *viññāṇaṃ*, Bodhi 2000, SN 22. 48). We have used the term thoughts instead of volitional.

5. Pali: *saṃkhittena pañcupādānakkhandhā dukkhā*.
6. (1) Eye and forms (objects), (2) ear and sounds, (3) nose and odours, (4) tongue and tastes, (5) body and tactile objects, (6) mind and mental phenomena (mind objects).
7. Pali: *diṭṭhe diṭṭhamattaṃ bhavissati, sute sutamattaṃ bhavissati, mute mutamattaṃ bhavissati, viññāte viññātamattaṃ bhavissatīti*.
8. *Vipassanā* is translated as inward vision, insight, intuition, introspection (Rhys Davids and Stede 1999). *Vipassanā* is derived from two roots. *Passanā* means seeing or perceiving. *Vi-* is a prefix with a complex.
9. 'This vast body of scriptures, recorded in the ancient Indian language known as Pali, is regarded by the Theravada school of Buddhism as the definitive recension of the Buddha-word, and among scholars it is generally considered our most reliable source for the original teachings of the historical Buddha Gotama' (Nanamoli and Bodhi 2001, Preface).
10. Pali: *Kiñca bhikkhave sabbaṃ: cakkhuñceva rūpā ca sotañca saddā ca ghāṇaṇca gandhā ca jivhā ca rasā ca kāyo ca phoṭṭhabbā ca mano ca dhammā ca idaṃ vuccati bhikkhave sabbaṃ*.
11. Pali: *imesaṃ tiṇṇaṃ dhammānaṃ saṅgati sannipāto samavāyo, ayaṃ vuccati*.
12. Pali: *Phuṭṭho, bhikkhave, vedeti, phuṭṭho ceteti, phuṭṭho sañjānāti*.
13. Pali: *Cakkhuñcāvuso, paṭicca rūpe ca uppajjati cakkhuvīññānaṃ, tiṇṇaṃ saṅgati phasso, phassapaccayā vedanā, yaṃ vedeti taṃ sañjānāti, yaṃ sañjānāti taṃ vitakketi, yaṃ vitakketi taṃ papañceti, yaṃ papañceti tatonidānaṃ purisaṃ papañcasaññāsañkhā samudācaranti atitānāgatapaccuppannesu cakkhuvīññeyyesu rūpesu*.
14. We have used the term mental proliferation to translate the word *papañca* so as to be consistent with the *sutta* text passages we have quoted from Bhikkhu Bodhi's translations.
15. 'Bhikkhus, there are taints that should be abandoned by seeing. There are taints that should be abandoned by restraining. There are taints that should be abandoned by using. There are taints that should be abandoned by enduring. There are taints that should be abandoned by avoiding. There are taints that should be abandoned by removing. There are taints that should be abandoned by developing' (Nanamoli and Bodhi 2001, MN 2). Three categories of taints (mental corruptions) are described in the *sutta* texts. Taints of sensual desire, taints of becoming or desire for future life, and taints of ignorance.
16. Pali: *Vedanā, saññā, cetanā, phasso, manasikāro—idaṃ vuccatāvuso, nāmaṃ*.
17. The body and the six senses are called the *rūpa*.
18. Pali: *Katamo ca, bhikkhave, dukkhassa nidānasambhavo? Taṇhā, bhikkhave, dukkhassa nidānasambhavo*.
19. 'With ignorance as a condition volitional formations (come to be); with volitional formation as condition, consciousness; with consciousness as condition, name-and-form; with name-and-form as condition, the six sense bases; with six sense bases as condition, contact; with contact as condition, feeling; with feeling as condition, craving; with craving as condition, clinging; with clinging as condition, existence; with existence as condition, birth; with birth as condition, ageing-and-death, sorrow, lamentation, pain, displeasure, and despair come to be. Such is the origin of this whole mass of suffering. This, bhikkhus, is called the dependent origination' (Bodhi 2000, SN, 12.1, 12.2).
20. Pali: *Evametassa kevalassa dukkhakkhandhassa samudayo hoti*.
21. Material or substratum by means of which an active process is kept alive or going, fuel, supply, provision (Rhys Davids and Stede, 1999).

22. Pali: *1.Kabaliṅkāro, 2. Phasso, 3. Manosañcetanā, 4. Viññāṇaṃ.*
23. Pali: *viññāṇaṭṭhitiyo.*
24. Pali: *Saṅkhārā.*
25. Pali: *Yadaniccaṃ taṃ dukkhaṃ.*
26. Pali: *yathābhūtaṃ.*
27. Pali: *satisampajaññāya.*
28. Pali: *Cakkhusamphasso, bhikkhave, anicco vipariṇāmi aññathābhāvī.*
29. Pali: *'nippapañcārāmassāvuso, sā niṭṭhā nippapañcaratino, na sā niṭṭhā papañcārāmassa papañcaratino'ti.*
30. Pali: *Chayime, bhikkhave, phassāyatanā adantā aguttā arakkhitā asaṃvutā dukkhādhivāhā honti.*

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